

Upcoming:

- **HW9 – Due Today**
- **Module 5 Videos – Due Sunday, April 6 – 11:59 PM**
- **NO CLASS Tuesday, April 8– Honors Day**
- **Watch Videos, Module 5 Postlab, Quiz 5 → Thursday April 10**
- **Exam 2 → Thursday, April 17**

Upcoming Due Dates

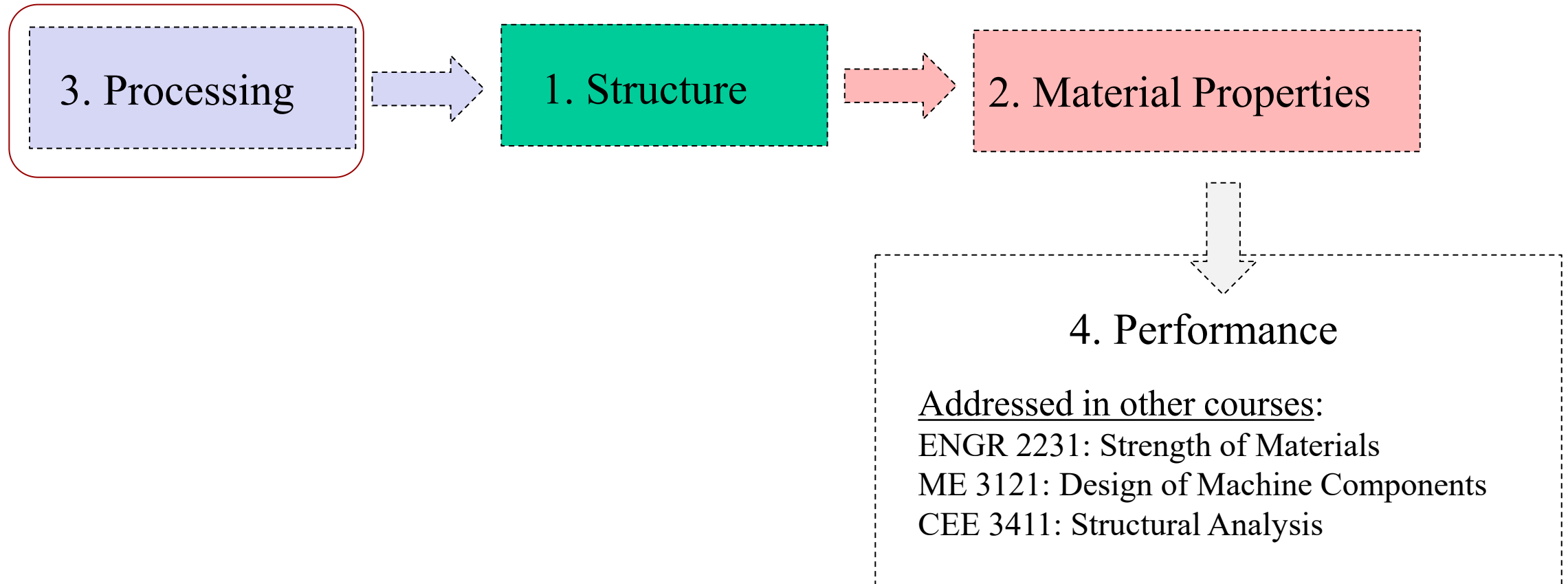
- HW9 – Mod 5 Prelab Due Today
– Rolling must be completed by Wednesday Night
- Module 4 video – Due Sunday Mar. 29
- Today: Sign up for Module 5 lab
- HW 10 - Due Tuesday, Mar 31
- Exam 2: Tuesday, April 2

Chapter 11:

Strengthening and Toughening

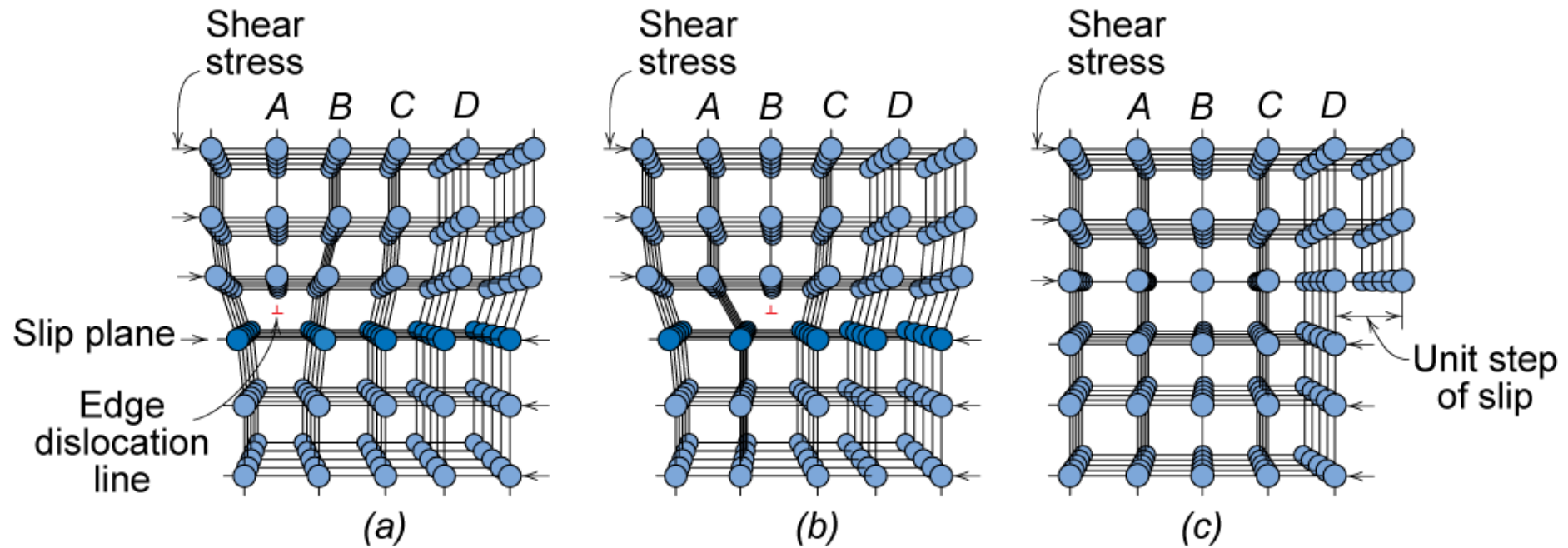
Today's Lecture

We are here!



REVIEW: Dislocation Motion (Metals)

Dislocation motion & plastic deformation



Adapted from Fig. 8.1,
Callister & Rethwisch 3e.

REVIEW: Crack Propagation

Cracks propagate due to sharpness of crack tip



Strategies for Strengthening Metals:

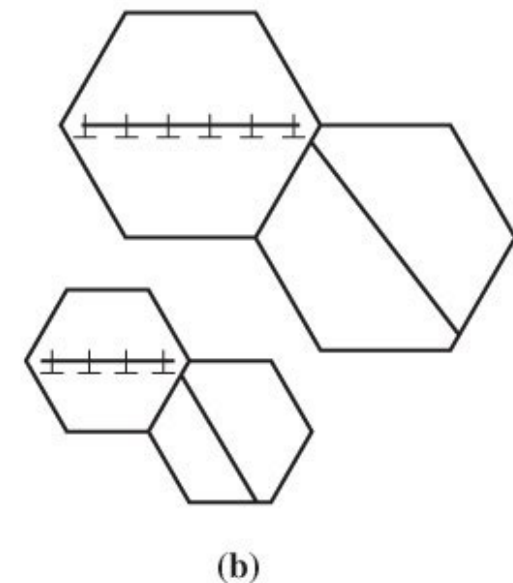
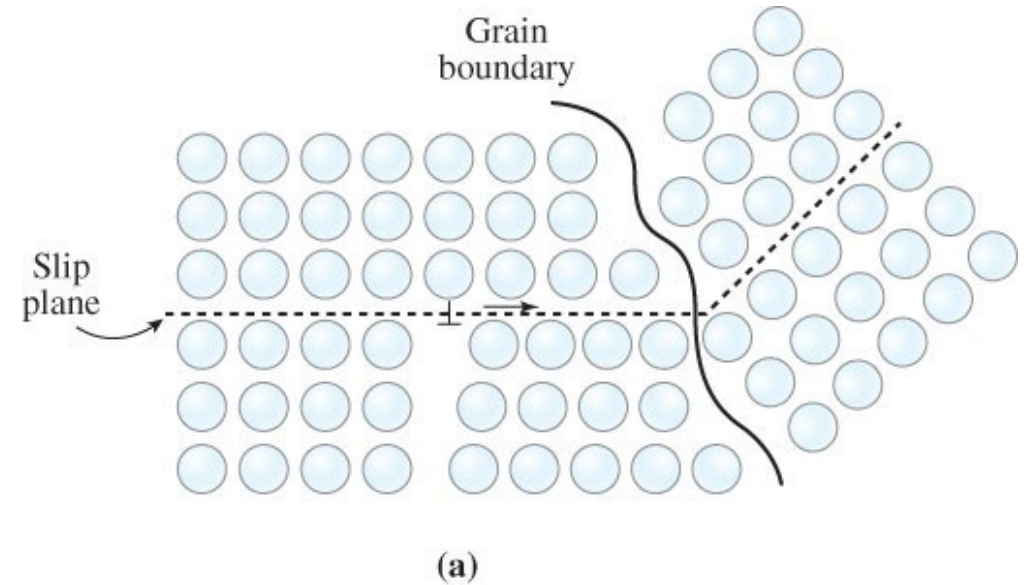
Add obstacles to block dislocation motion

1: Reduce Grain Size

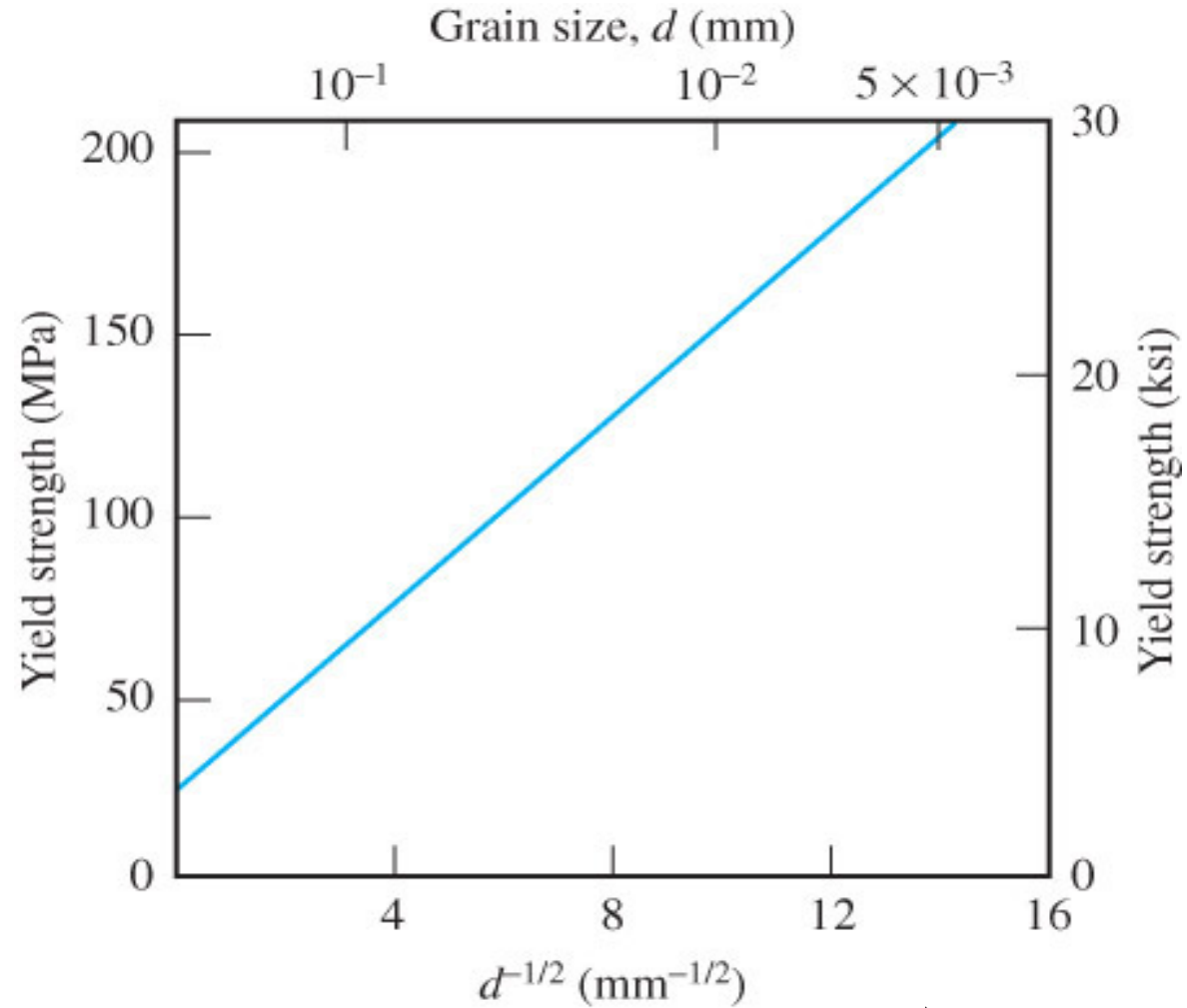
Grain boundaries are barriers to slip.

- Barrier "strength" increases with increasing angle of misorientation.
- Smaller grain size → more barriers to slip.
- Hall-Petch Equation: $\sigma_{yield} = \sigma_o + k_y d^{-1/2}$

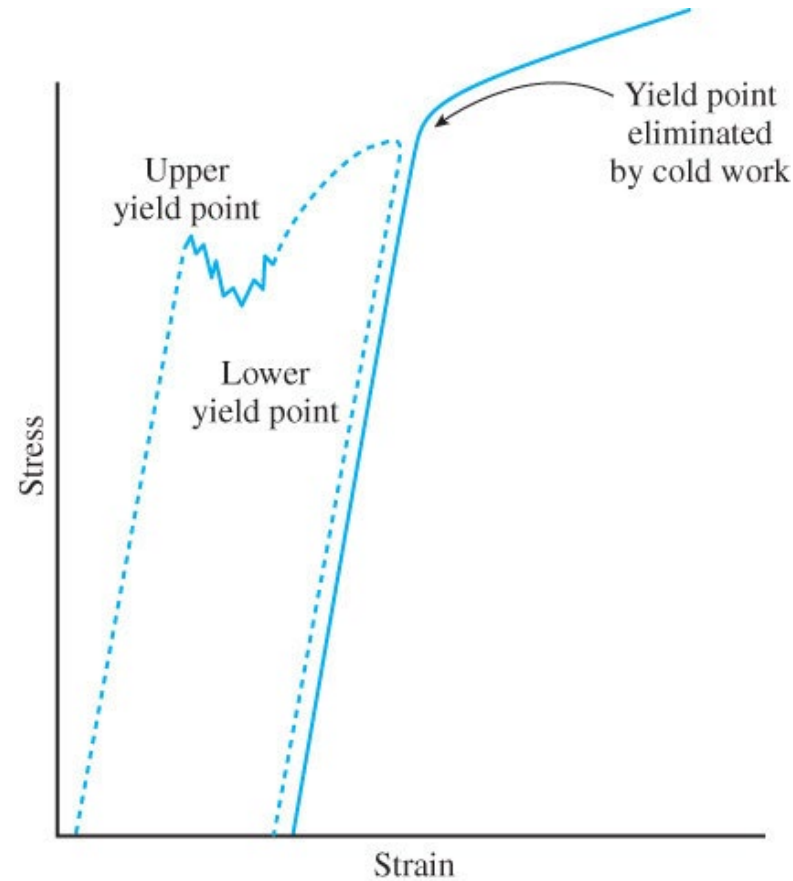
d ~ average grain diameter



Hall-Petch Relationship



2. Strengthening Metals by Cold Work: Strain Hardening



Working (i.e., plastically deforming) a metal below its recrystallization temperature results in strain hardening, as shown.

2. Strategies for Strengthening Metals: Cold Work (%CW)

- Room temperature deformation.
- Common forming operations (next 4 slides) change the cross sectional area
- %CW is calculated as:

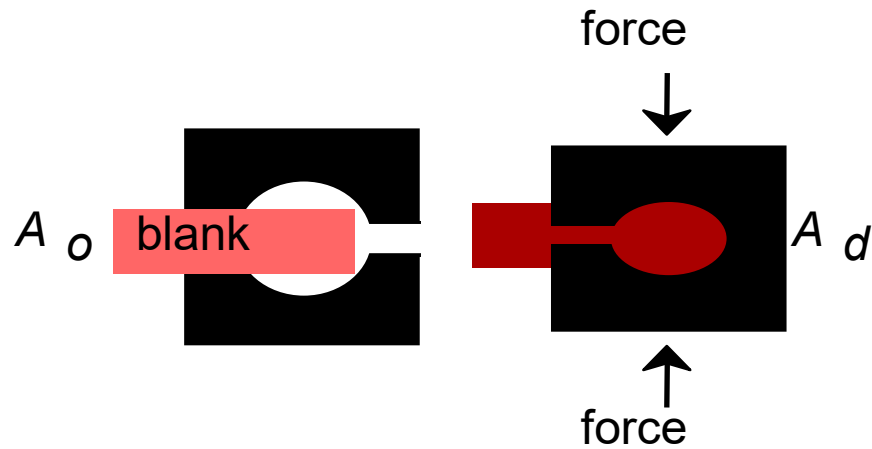
$$\%CW = \frac{A_o - A_d}{A_o}$$

A_o – original cross-section area

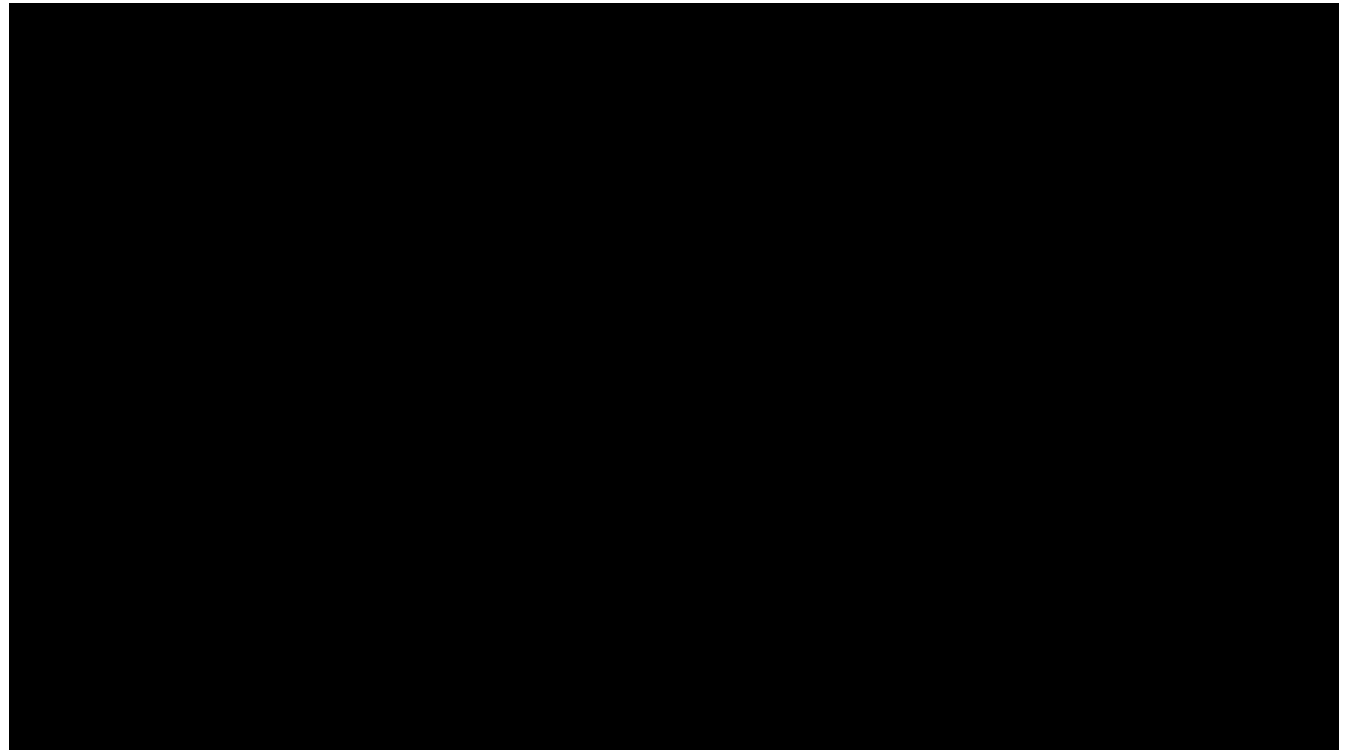
A_d – deformed cross-section area

Cold Working: Forging

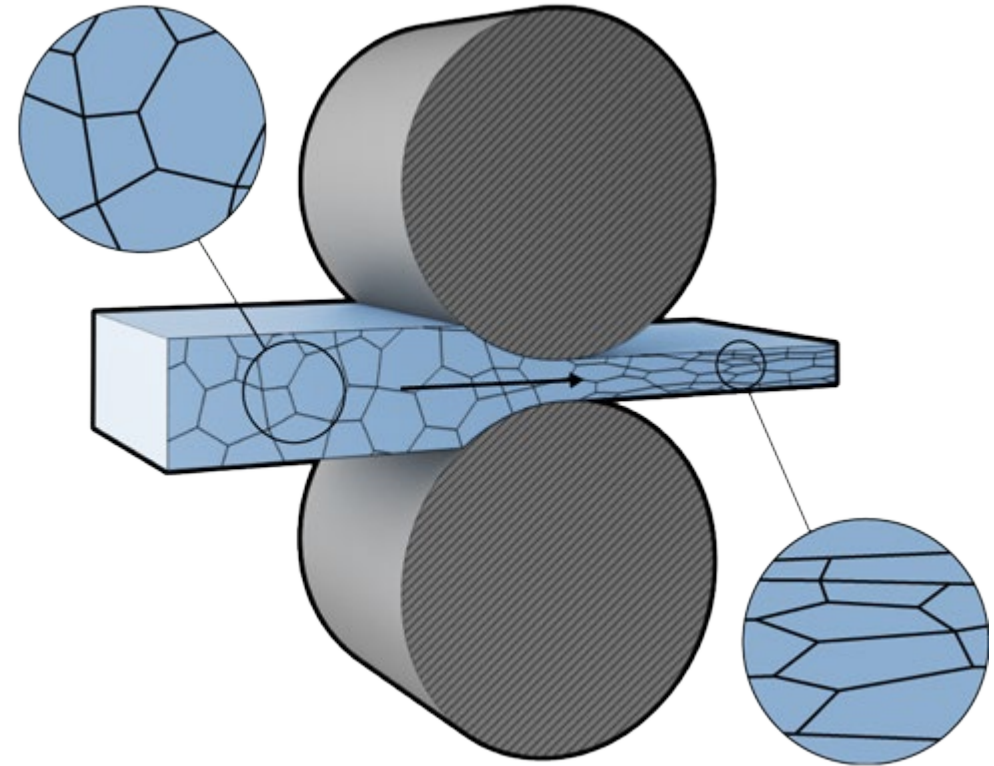
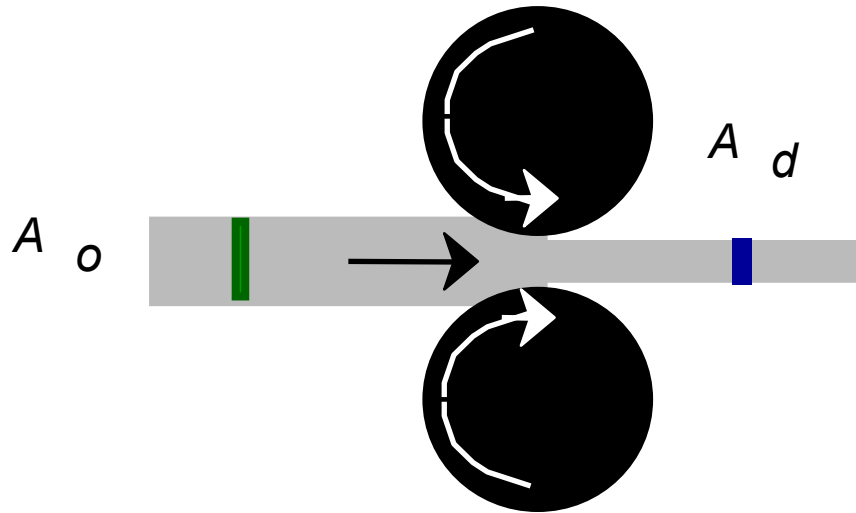
A metalworking process that shapes metals parts at near room temperature by applying compressive forces between dies. Often involves hammering/pounding of the workpiece.



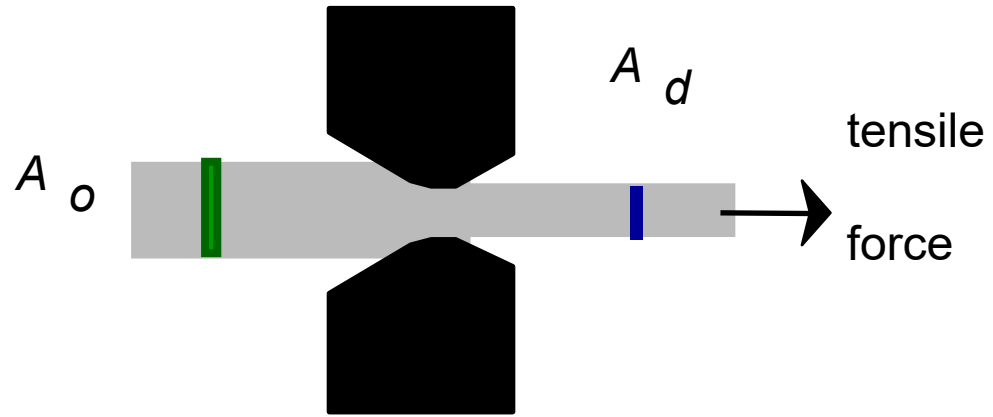
Adapted from Fig. 14.2, *Callister & Rethwisch 3e*.



Cold Working: Rolling

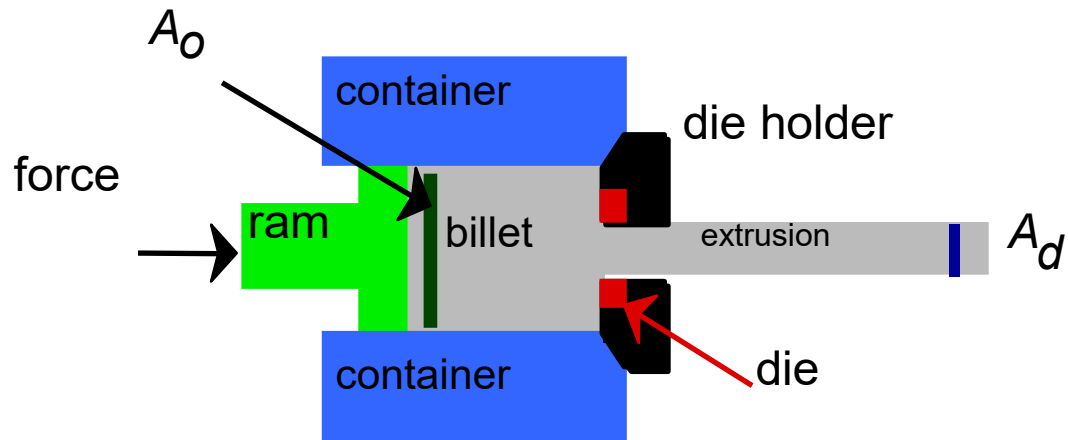


Cold Working: Drawing



<https://www.youtube.com/watch?v=QKAg1yMZIpY>

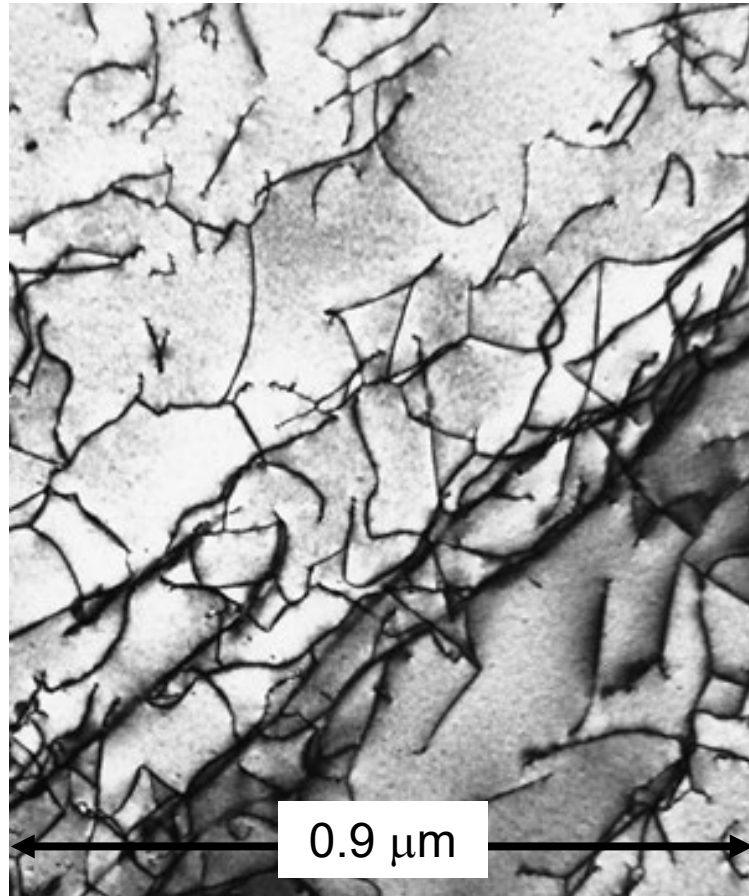
Cold Working: Extrusion



<https://www.youtube.com/watch?v=QKAg1yMZIpY>

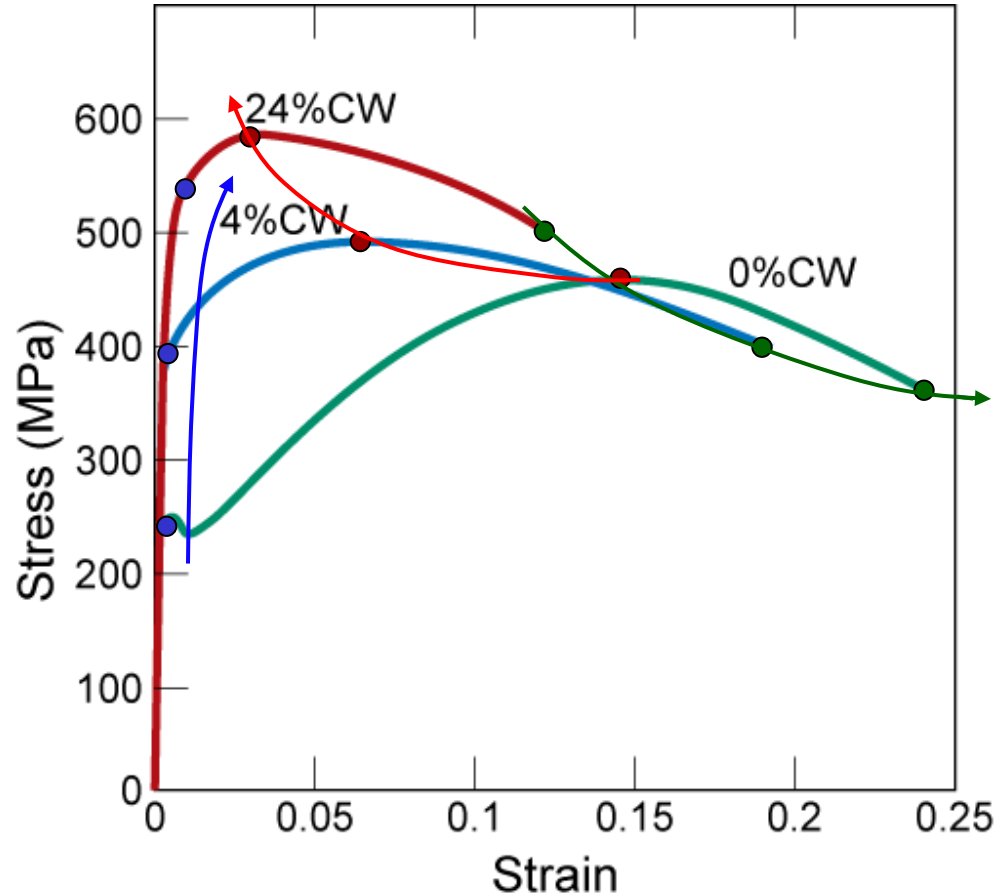
Dislocations During Cold Work

Ti alloy after cold working:



- Dislocation number increases and dislocations entangle with one another during cold work.
- Dislocation motion becomes more difficult.

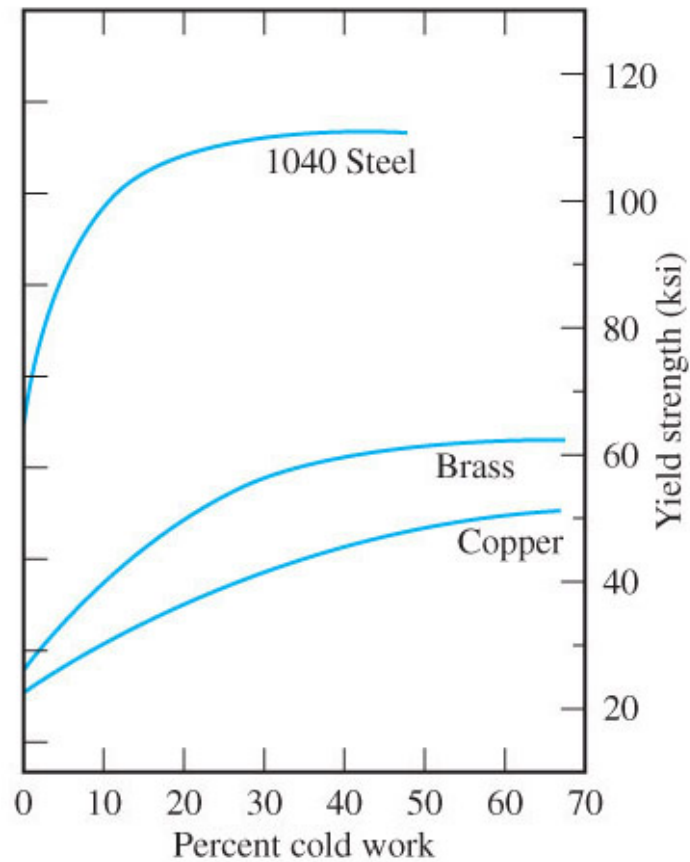
Impact of Cold Work



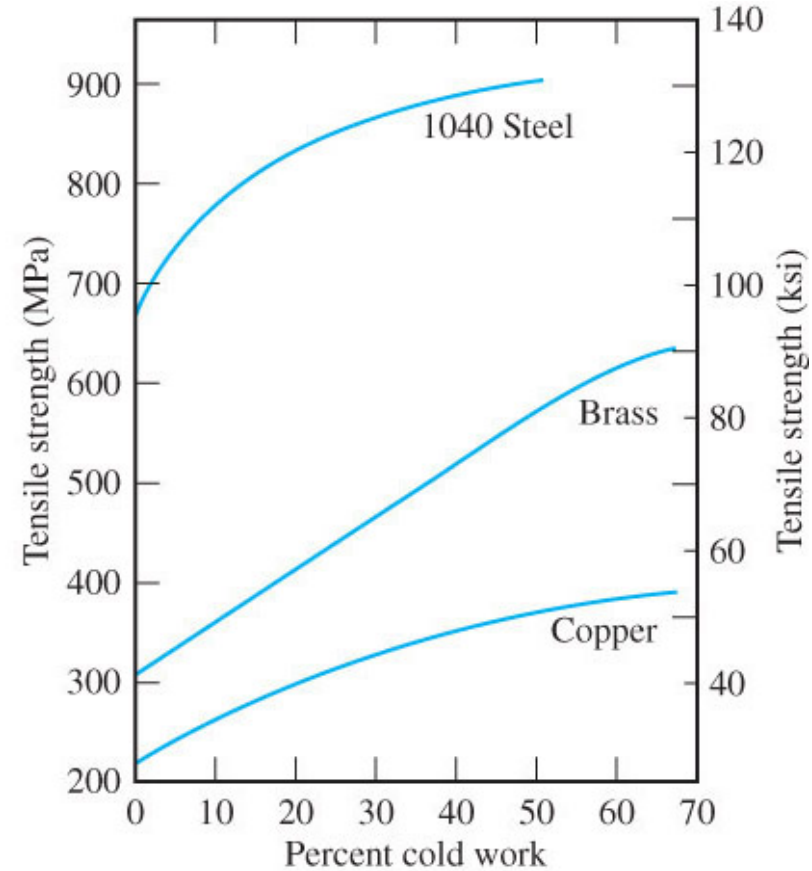
As cold work is increased:

- Yield strength (σ_y) increases.
- Tensile strength (TS) increases.
- Ductility ($\%EL$ or $\%AR$) decreases.

Strengthening Metals by Cold Work: **Strain Hardening**

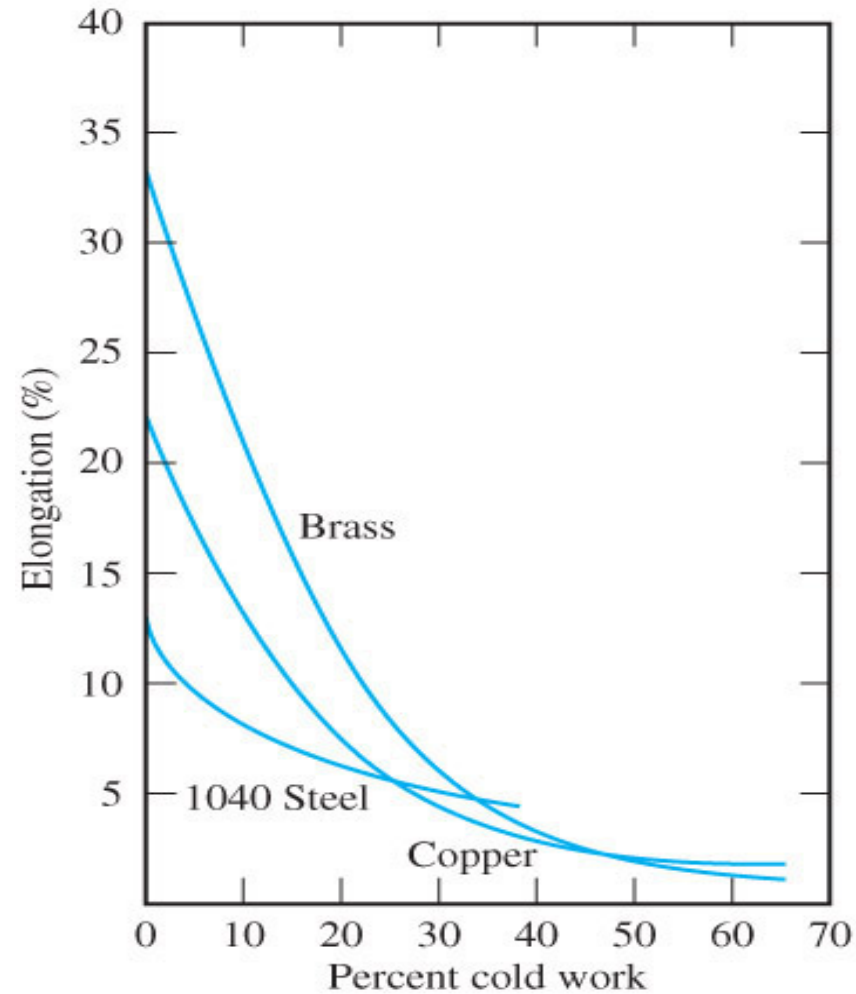


(a)



(b)

Strengthening Metals by Cold Work: Strain Hardening

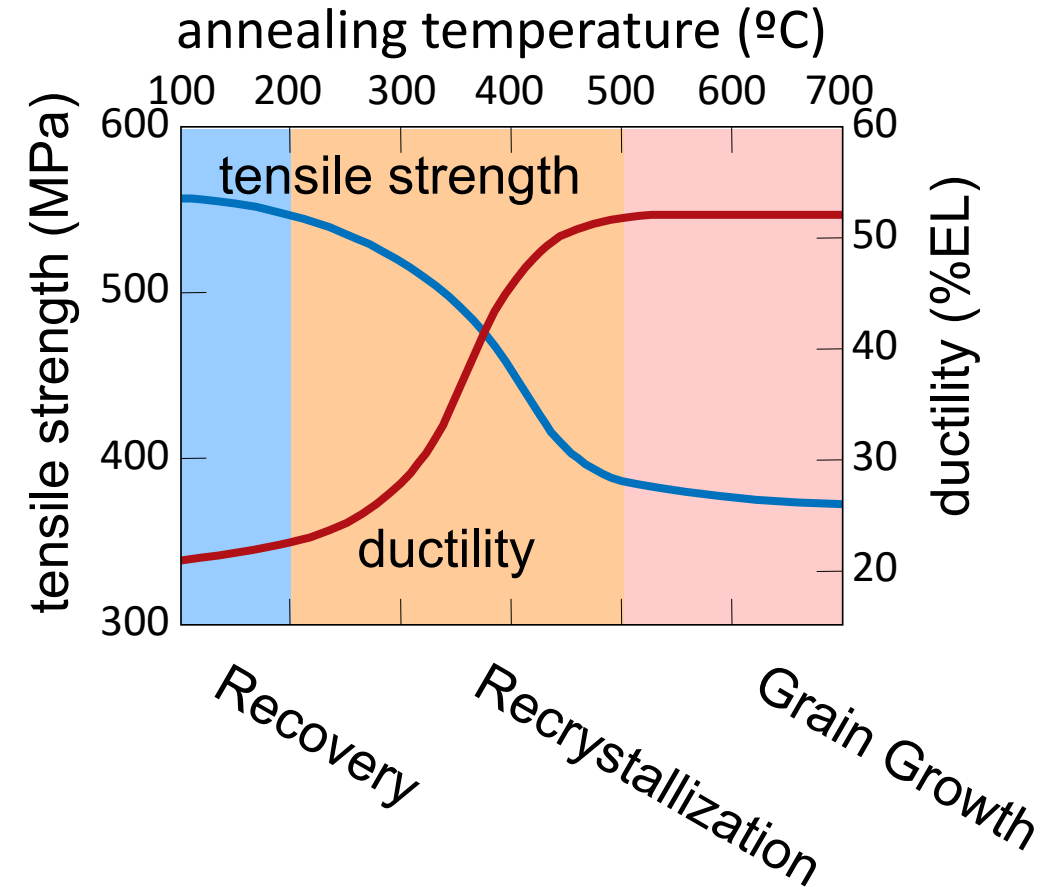


(c)

Effect of Heating After %CW

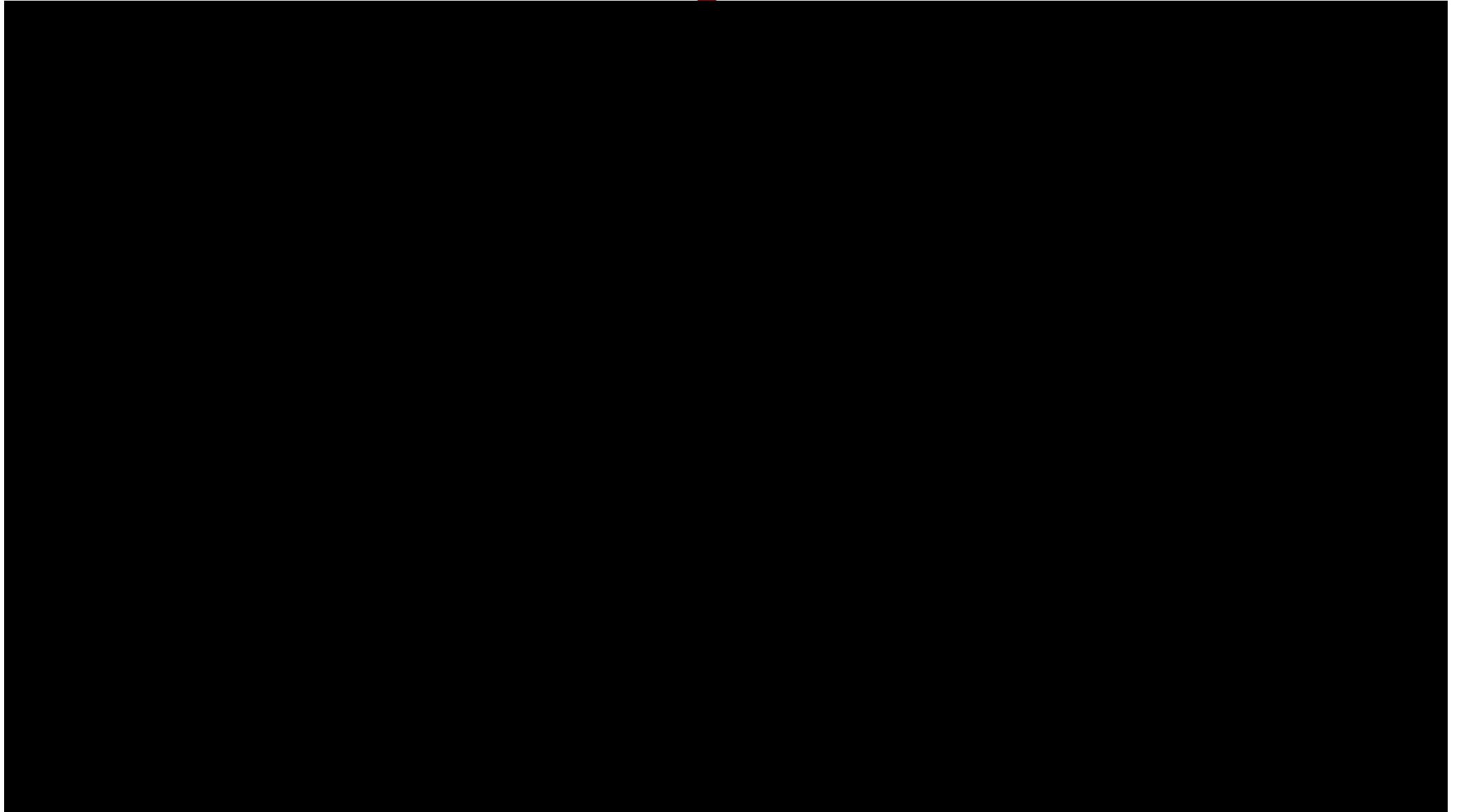
Annealing:

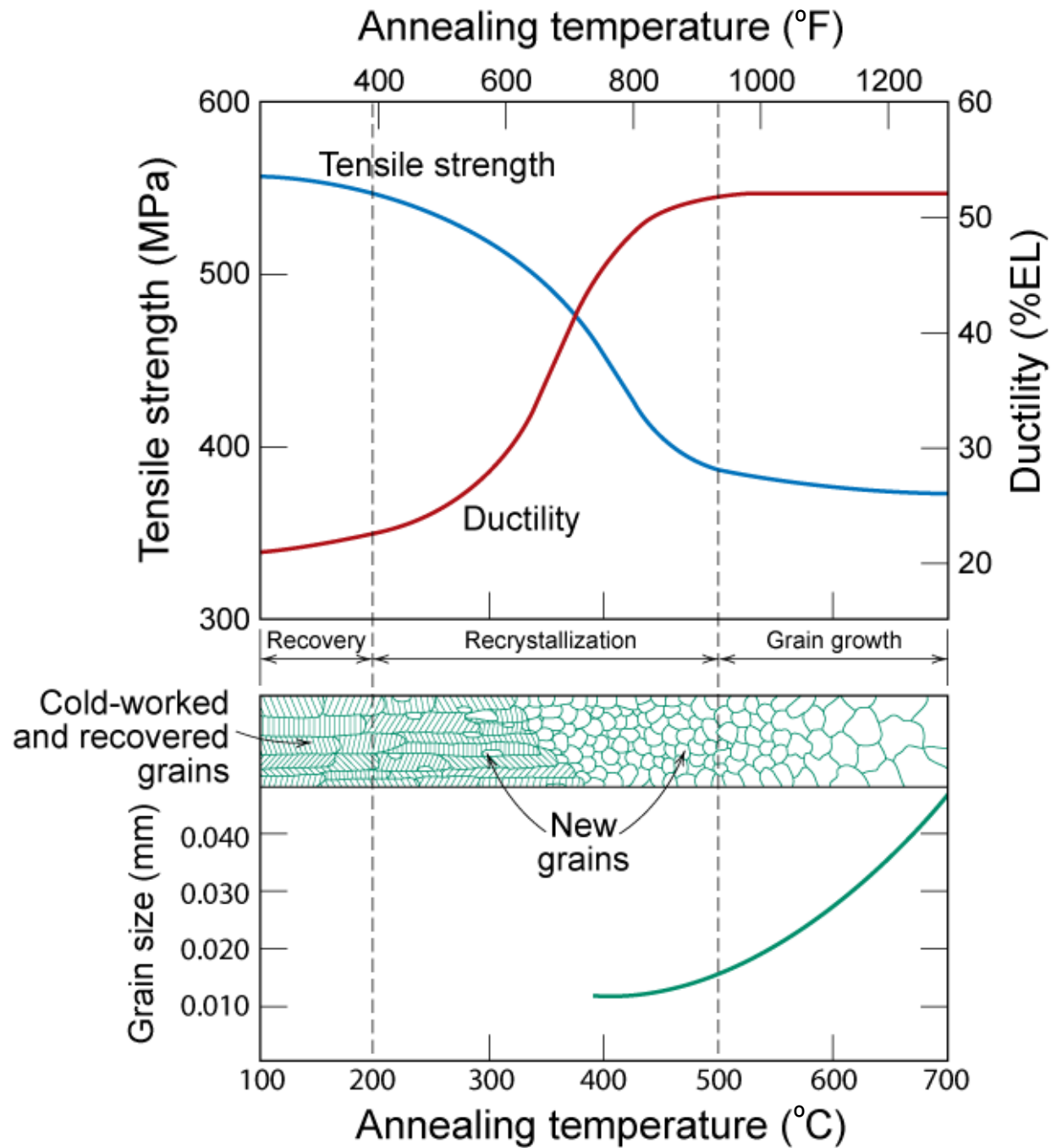
- Involves heating the material past the recrystallization temp
- 1 hour treatment at T_{anneal} ...
 - decreases TS
 - increases % EL .
- Effects of cold work are reversed!



Adapted from Fig. 8.22, *Callister & Rethwisch 3e*. (Fig. 8.22 is adapted from G. Sachs and K.R. van Horn, *Practical Metallurgy, Applied Metallurgy, and the Industrial Processing of Ferrous and Nonferrous Metals and Alloys*, American Society for Metals, 1940, p. 139.)

Annealing Process



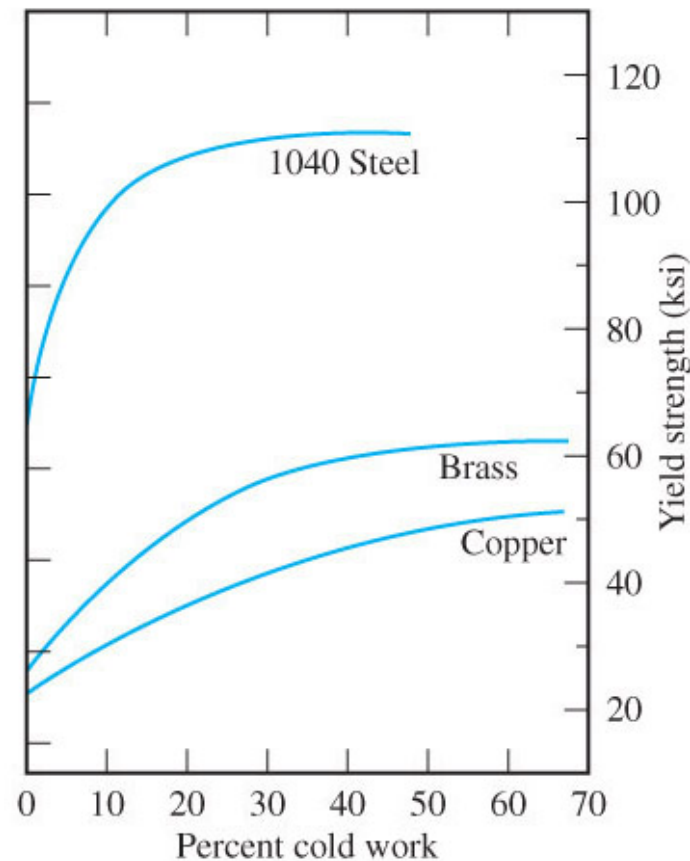


T_R = recrystallization temperature

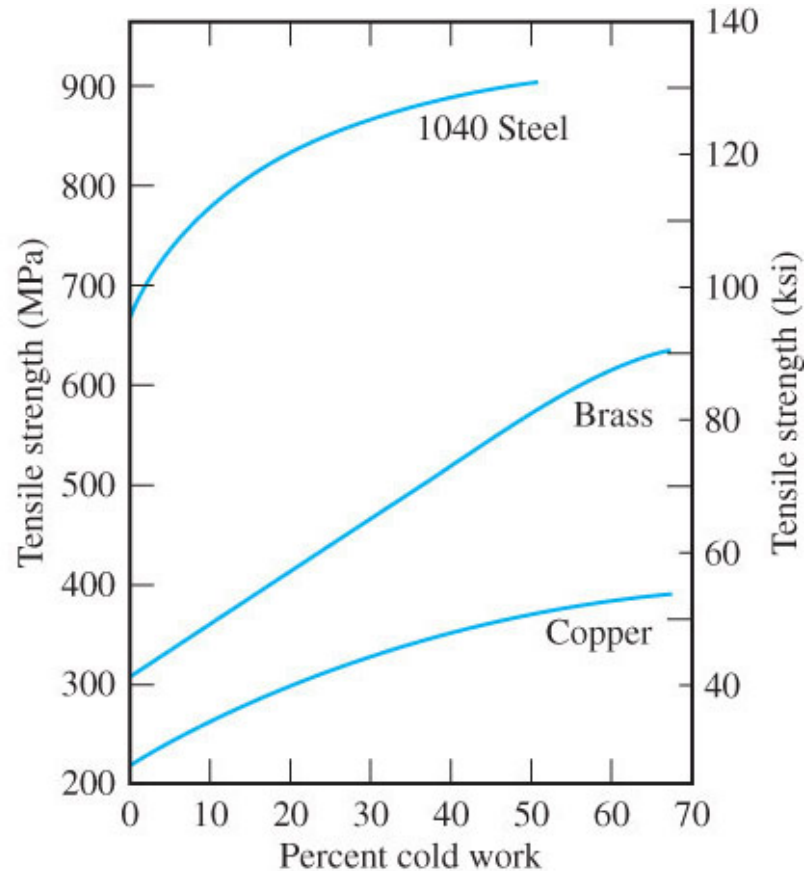
Adapted from Fig. 8.22,
Callister & Rethwisch 3e.

Example:

For a cold-worked cylindrical copper specimen with tensile strength of 300 MPa, the cold-worked radius is 0.3 cm. What is the radius before deformation?



(a)

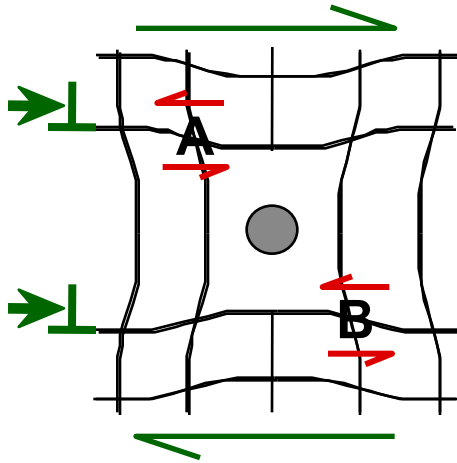


(b)

3. Solid Solutions: Strengthening by alloys

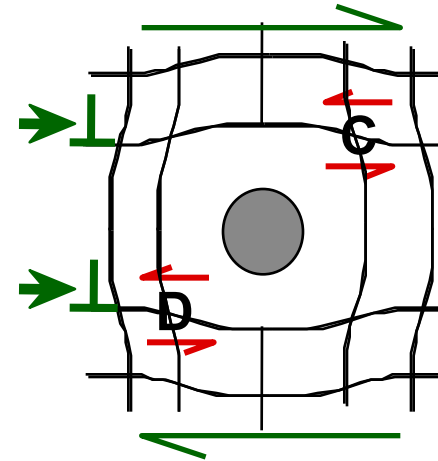
- Impurity atoms distort the lattice & generate stress.
- Stress can produce a barrier to dislocation motion.

Smaller substitutional impurity



Impurity generates local stress at **A** and **B** that opposes dislocation motion to the right.

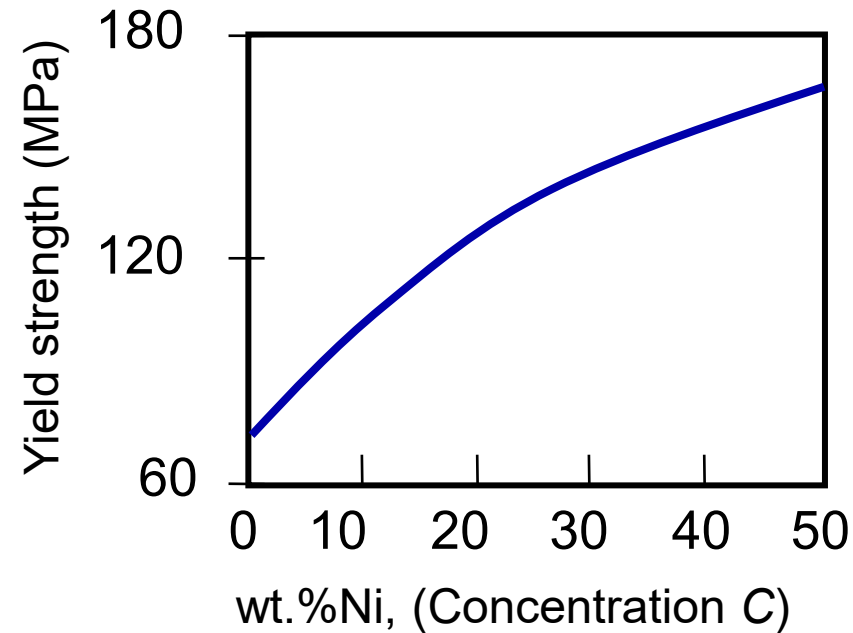
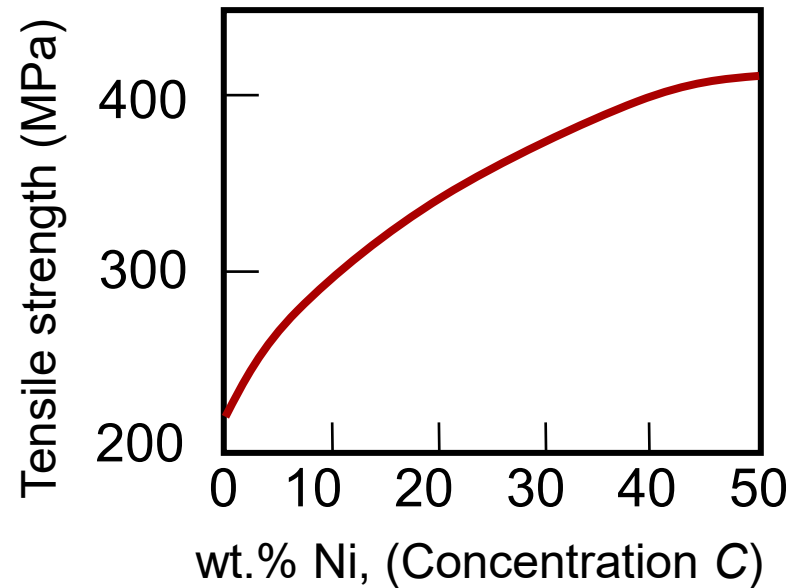
Larger substitutional impurity



Impurity generates local stress at **C** and **D** that opposes dislocation motion to the right.

Ex: Strengthening in Copper

- Tensile strength & yield strength increase with wt% Ni.

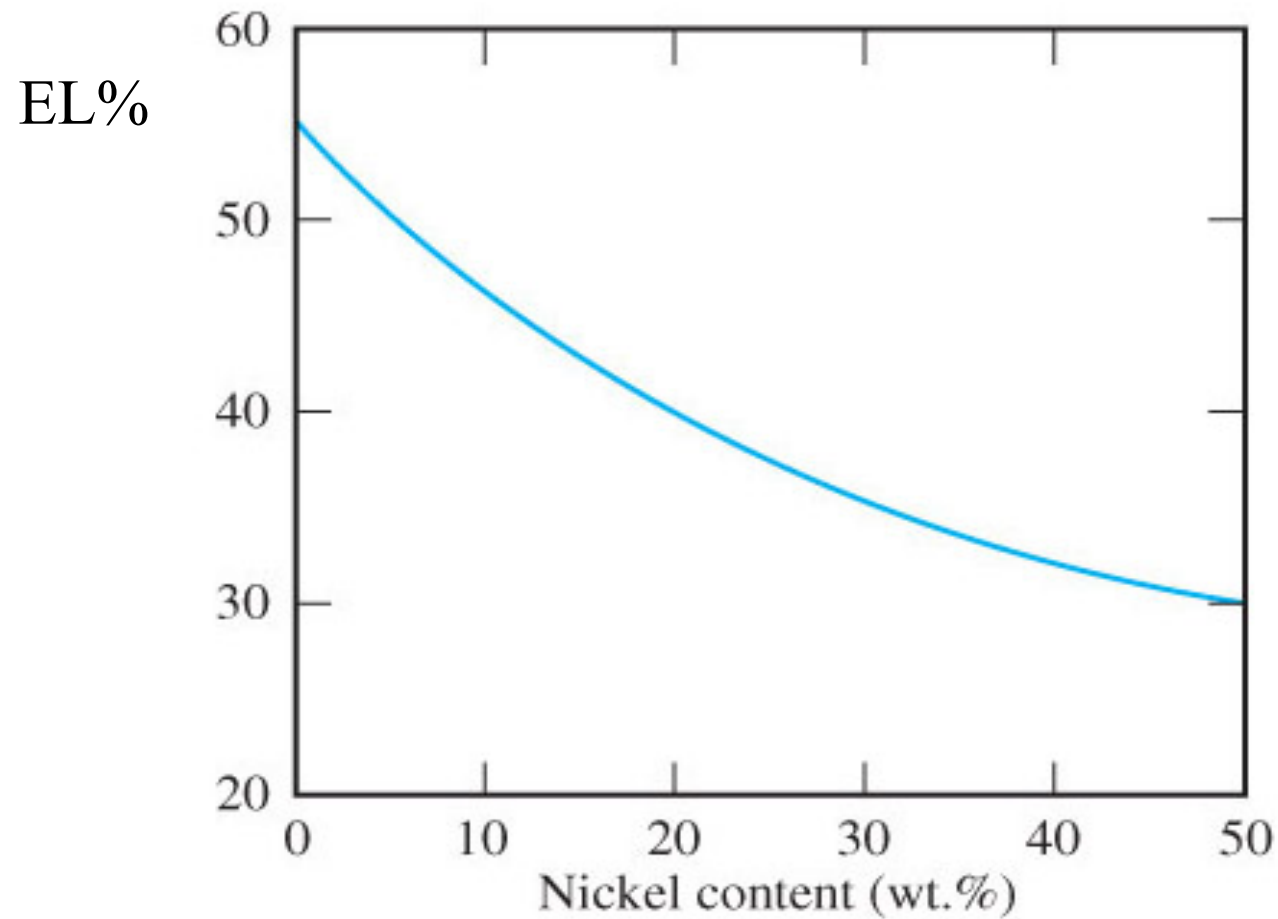


- Empirical relation: $\sigma_y \sim C^{1/2}$
- Alloying increases σ_y and TS .

Adapted from Fig. 8.16 (a) and (b), *Callister & Rethwisch 3e*.

Strengthening Metals by Alloying:

Ductility decreases



(c)

Metals

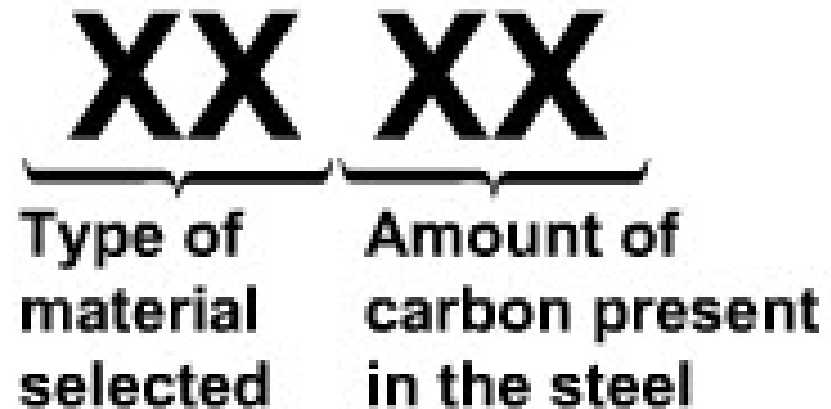
- **Engineering metals:** mechanical strength is needed
 - Most widely used engineering metal—iron
 - Iron based alloys---steels
- **Metal alloys:** combination of two or more chemical elements where metal is the major constituent.

Irons and steels

- The strength of pure iron is low
- Cast irons and Steels are iron-based alloys
- Addition of a small amount of carbon can strengthen the material greatly

Naming systems for irons and steels

AISI / SAE Steel Designation System



Example: AISI 1020

AISI 1020

- **First two digits indicates the alloy content other than carbon. For example, 10 – plain carbon.**

SAE designation	Type
1xxx	Carbon steels
2xxx	Nickel steels
3xxx	Nickel-chromium steels
4xxx	Molybdenum steels
5xxx	Chromium steels
6xxx	Chromium-vanadium steels
7xxx	Tungsten steels
8xxx	Nickel-chromium-vanadium steels
9xxx	Silicon-manganese steels

Useful Reference:

<https://www.engineersedge.com/materials/aisi-sae-steel-identification.htm>

<https://www.theprocesspiping.com/wp-content/uploads/2017/12/sae-aisi-first-digit.png>

AISI 1020

- First two digits indicates the alloy content other than carbon. For example, 10 – plain carbon.
- **Last two digits show the carbon content in hundredths of a percent.**

AISI 1020: 0.20% Carbon

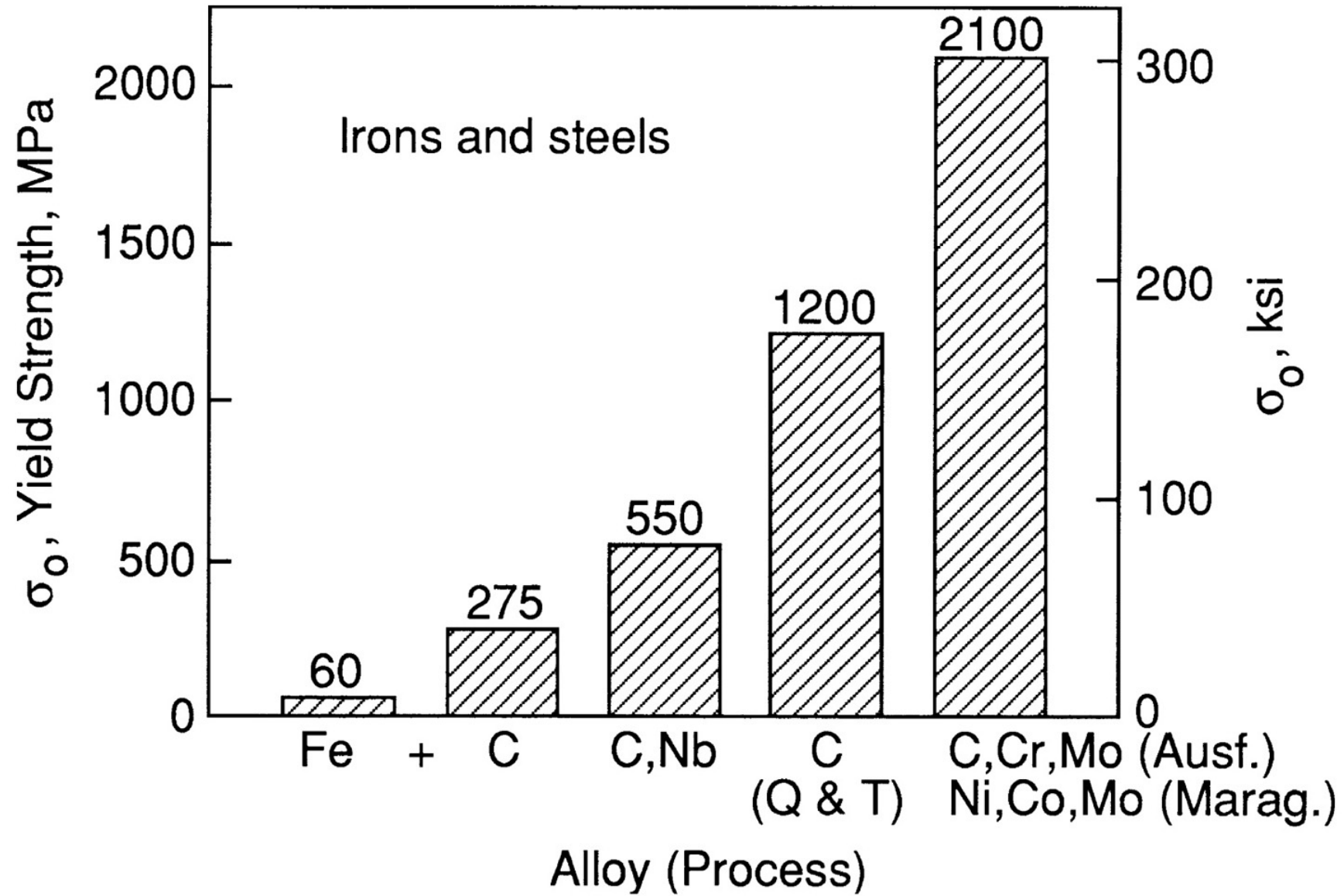
- low carbon steel ($<0.25\%$)

AISI 1045: 0.45% Carbon

- Medium carbon steel ($0.3\% \sim 0.6\%$)

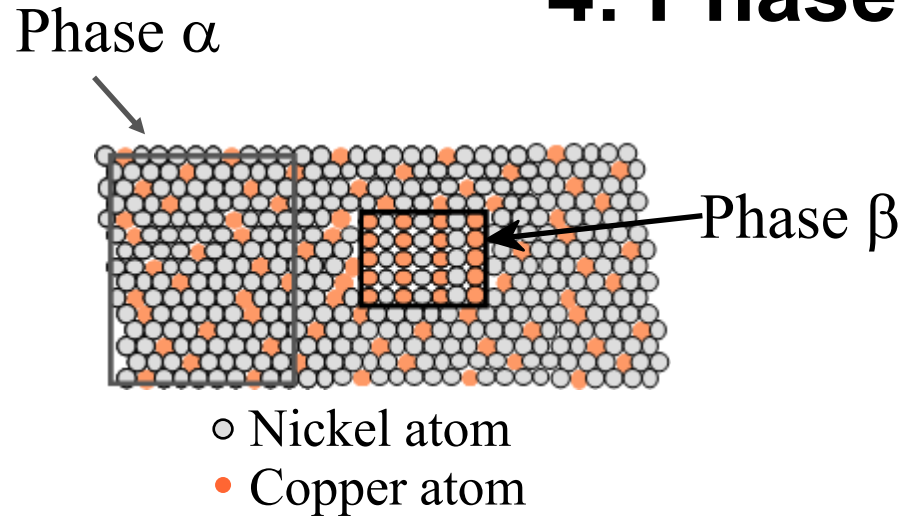
AISI 1095: 0.95% Carbon

- High carbon steel ($0.7\% \sim 1\%$)



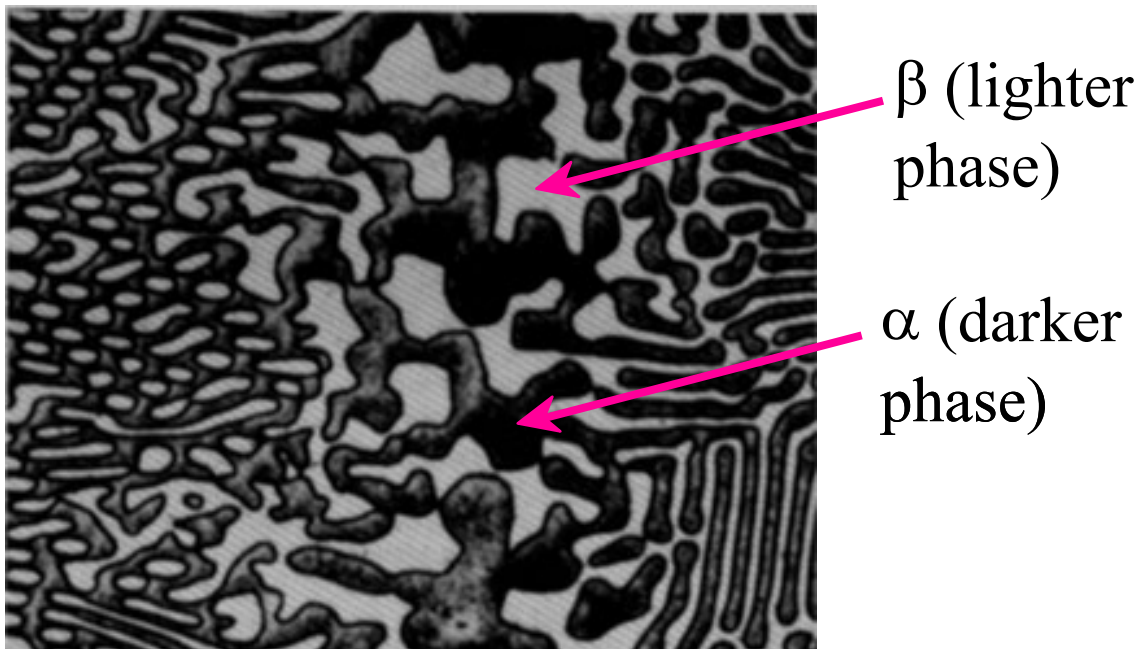
Strategies for Strengthening Metals:

4: Phase Boundary Strengthening



Phase boundaries

- Def: the junction of two crystals which differ in lattice structure and composition
- These different crystals are called phases
- Phases are formed during the solidification of an alloy

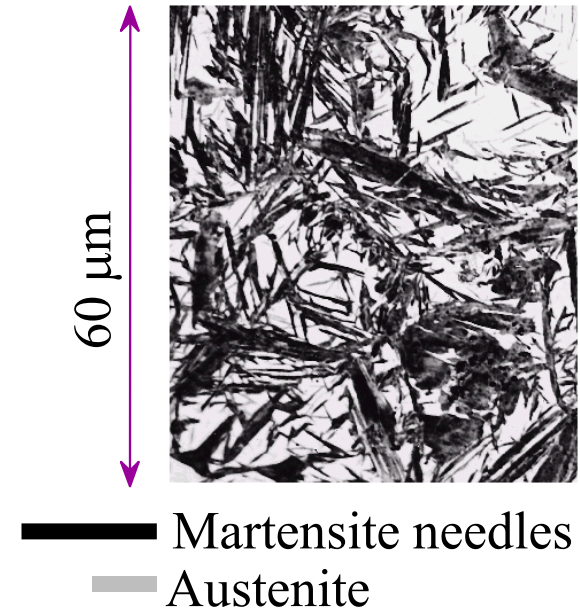
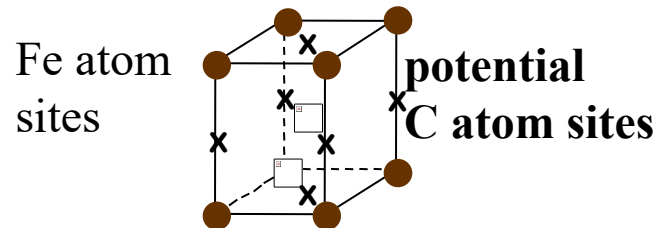


Strategies for Strengthening Metals:

5: Quenching

- Martensite:
-- γ (FCC) to Martensite (BCT)

(involves single atom jumps)



<https://www.youtube.com/watch?v=gPKkgmDoEoU>

Heat Treatment and Quenching



Fracture: Crack Propagation

Cracks propagate due to sharpness of crack tip



Toughening: lower stress concentration at the tip of the crack

Toughening Ceramics

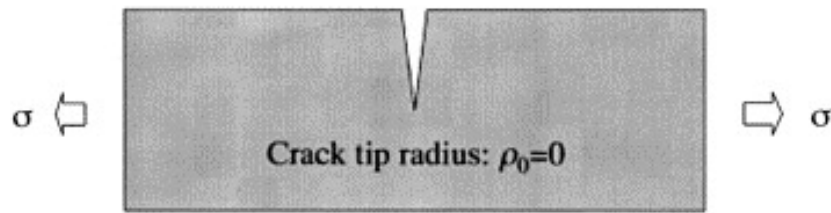
Ceramics: ductility and toughness are concerns

Strategies:

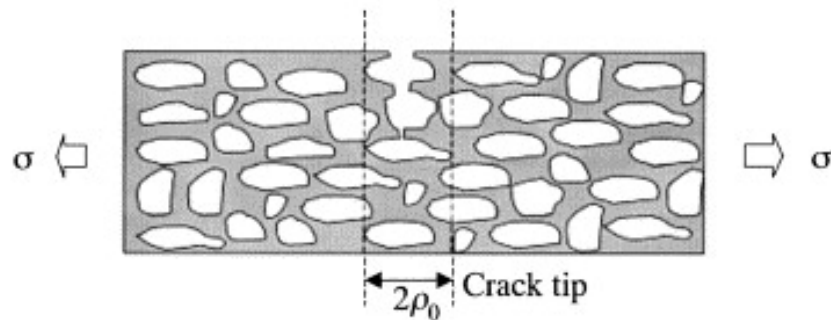
- Introduce microporosity to increase crack tip radius (blunting crack tip)
- Add ductile phase to arrest cracks
- Develop compressive residual stress at the crack tip

Toughening Ceramics

Introduce microporosity to increase crack tip radius (blunting crack tip)



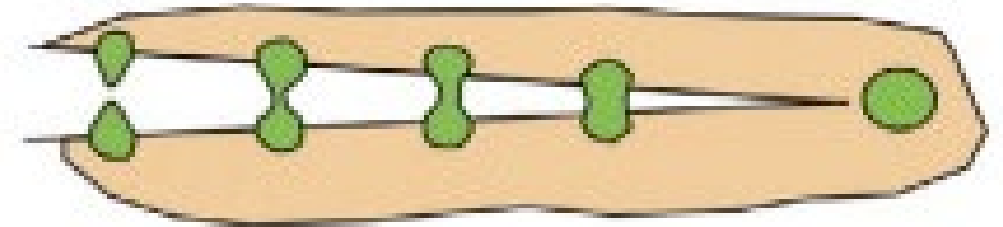
(a) Dense material



(b) Porous material

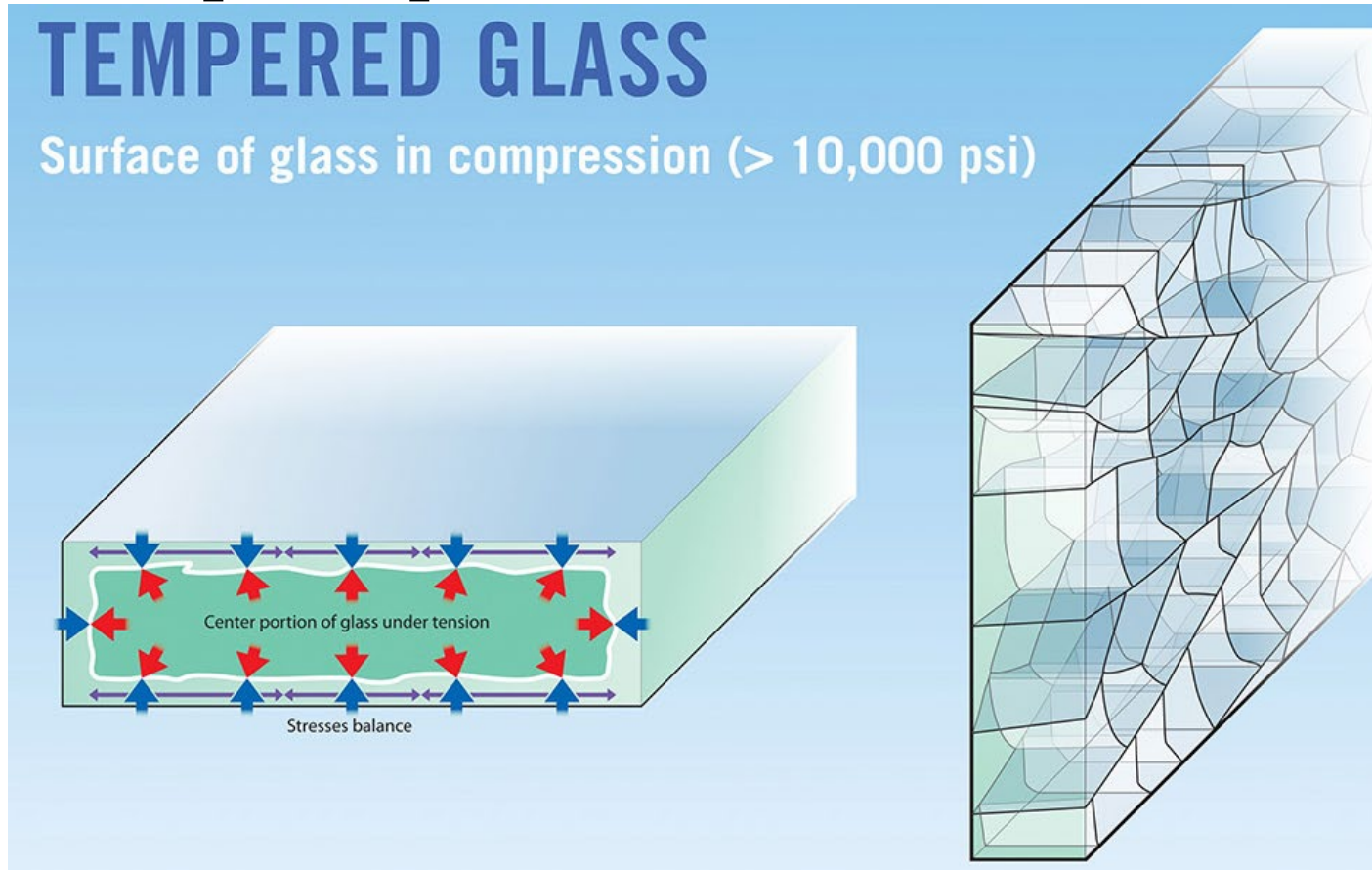
Add ductile phase to arrest cracks

Ductile phase bridging of a crack in a brittle matrix



Toughening Ceramics

Develop compressive residual stress at the crack tip



<https://glassed.vitroglazings.com/topics/heat-strengthened-vs-tempered-glass>

Strengthening polymers

- Increase Molecular Weight (long chain)
- Increase degree of crystallization (more aligned chains)

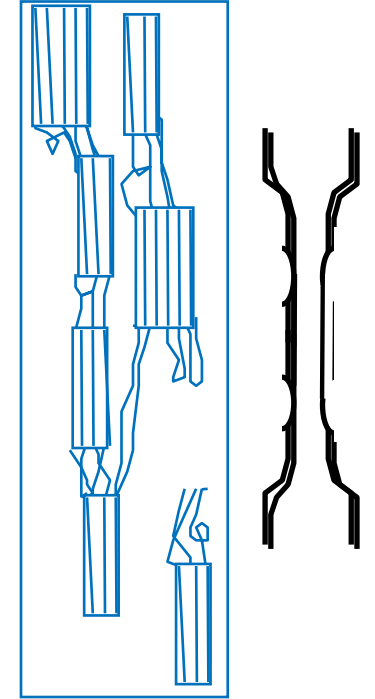
Drawing...(ex: monofilament fishline)

- stretches the polymer prior to use
- aligns chains in the stretching direction

Annealing

- increase E and TS , decrease $EL\%$ (opposite effects for metals)

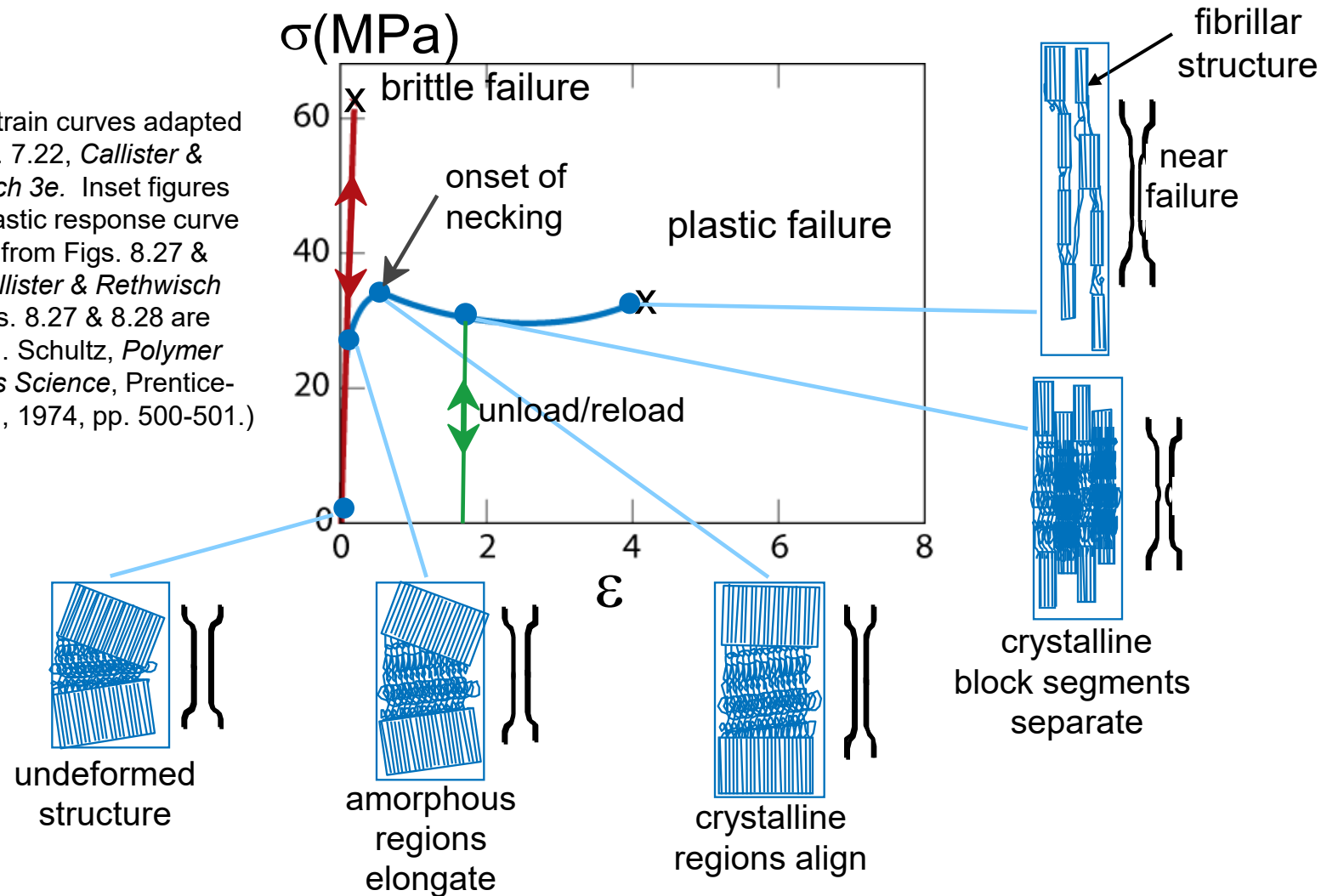
for semicrystalline polymers, annealing can lead to an increase in the percent crystallinity and in crystallite size and perfection, as well as modifications of then spherulite structure. Different mechanism from metal annealing

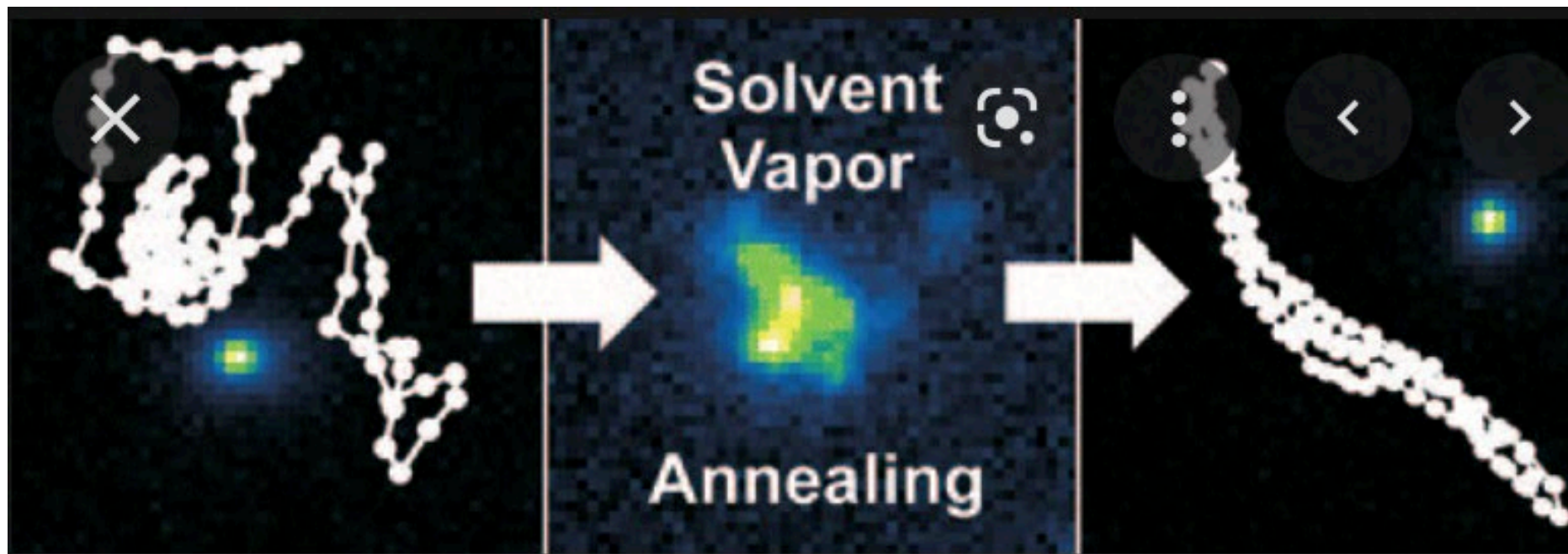


Strengthening for polymers:

Semicrystalline (Plastic) Polymers

Stress-strain curves adapted from Fig. 7.22, *Callister & Rethwisch 3e*. Inset figures along plastic response curve adapted from Figs. 8.27 & 8.28, *Callister & Rethwisch 3e*. (Figs. 8.27 & 8.28 are from J.M. Schultz, *Polymer Materials Science*, Prentice-Hall, Inc., 1974, pp. 500-501.)



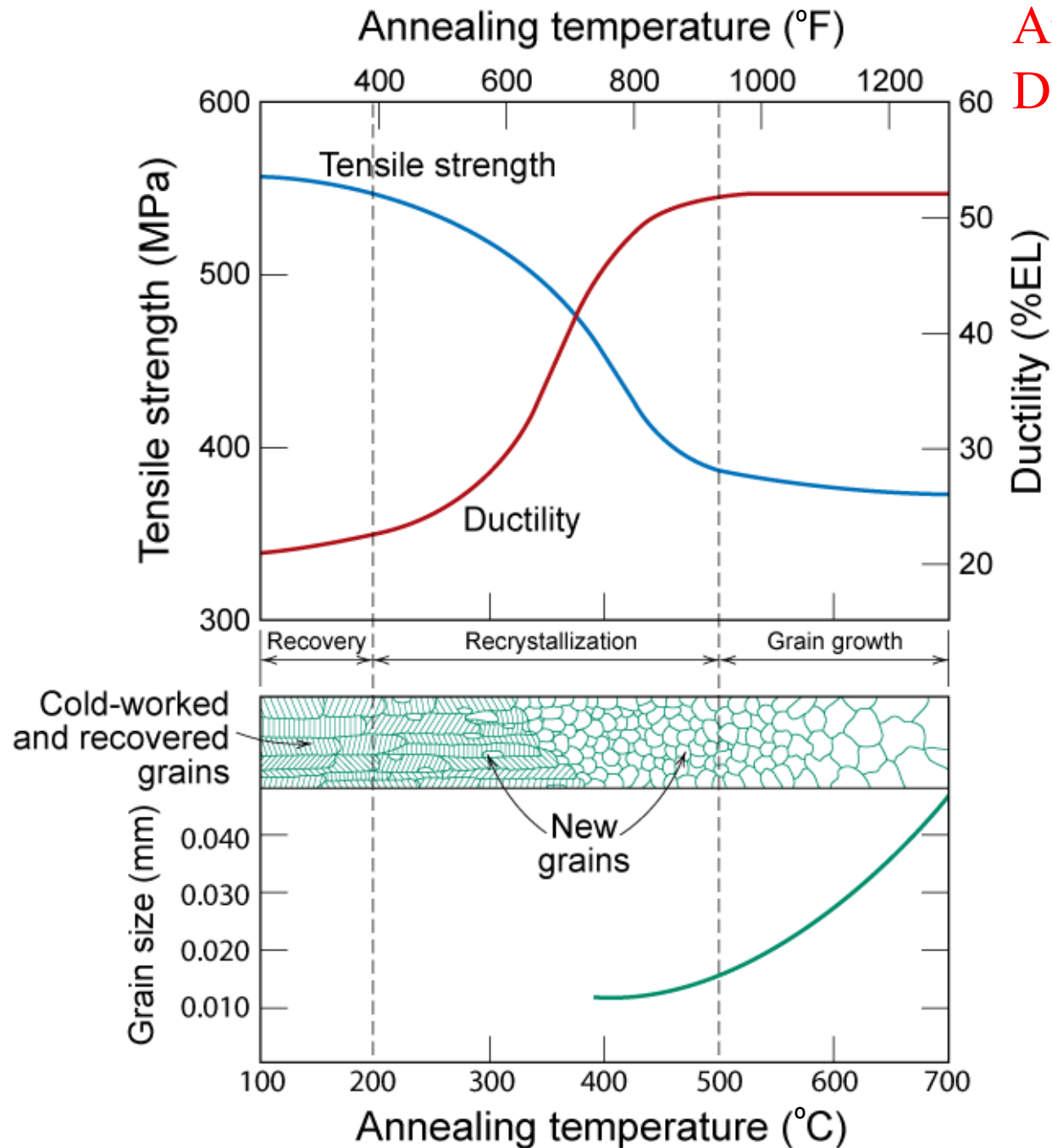


ChemistryViews
**Polymer Annealing One Chain at a Time -
ChemistryViews**
Visit

https://www.chemistryviews.org/details/ezone/1009209/Polymer_Annealing_One_Chain_at_a_Time/

Annealing of metals

Different from Polymer annealing



T_R = recrystallization temperature

Adapted from Fig. 8.22,
Callister & Rethwisch 3e.